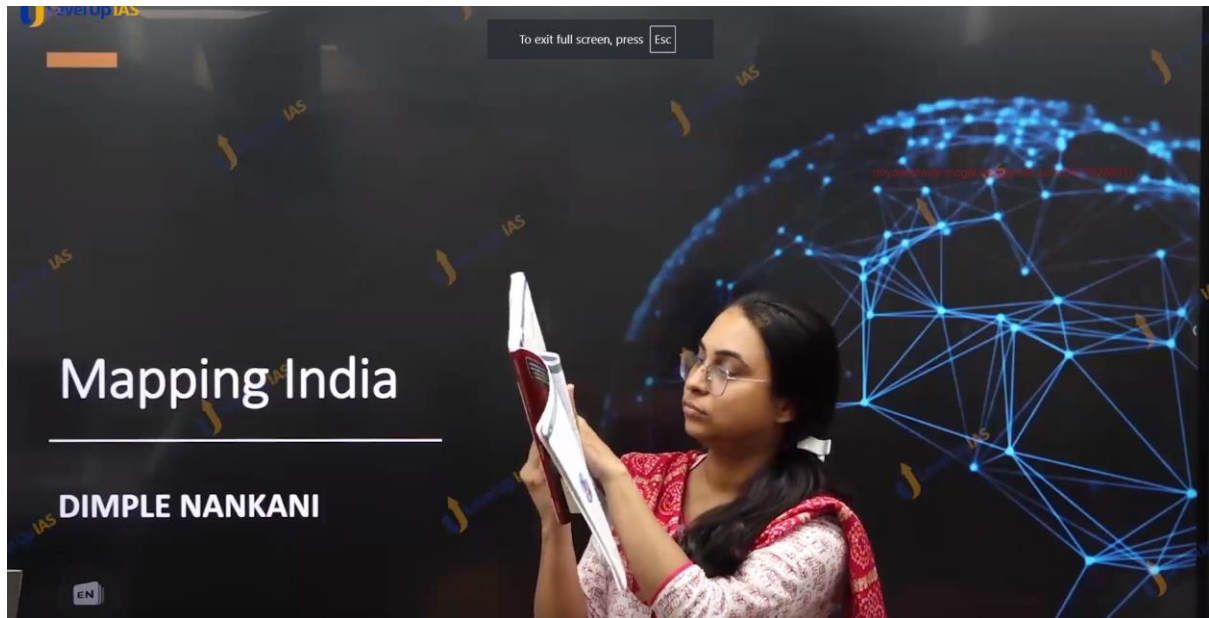


(28) Peninsular Plateau



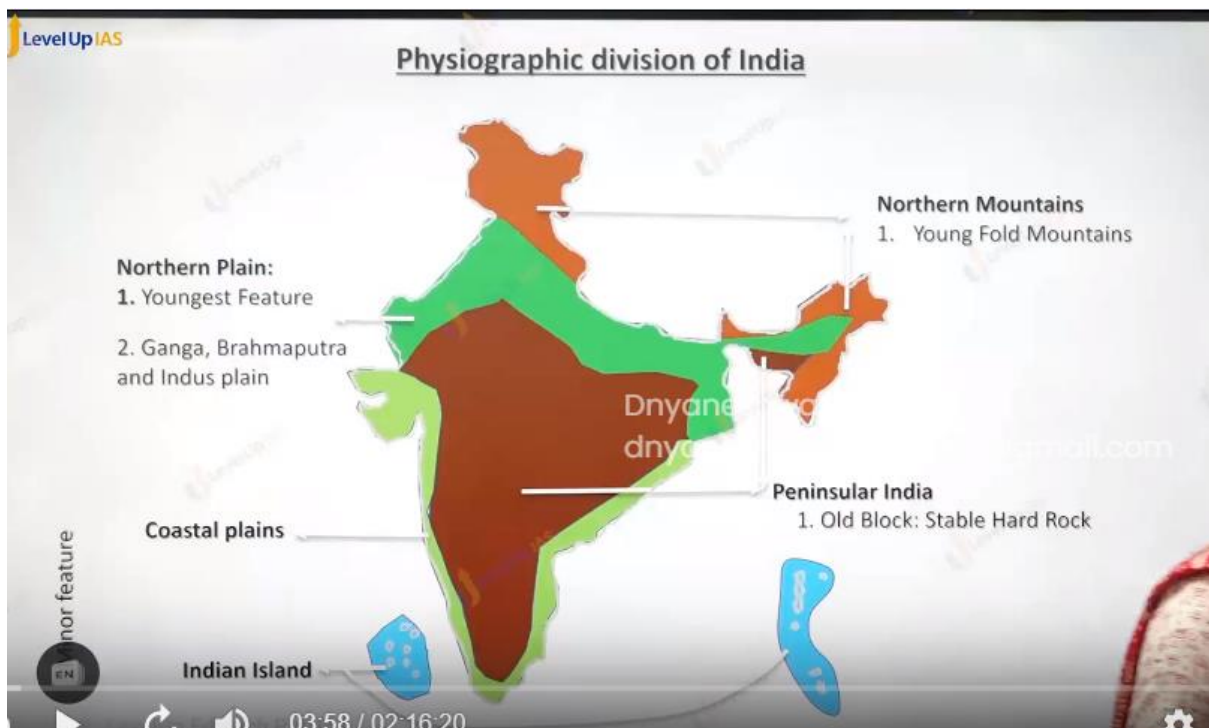
Mapping India:

-
1. Tides refers to vertical movement of water nearly each place on the earth experiences two high tide tow low tide
-

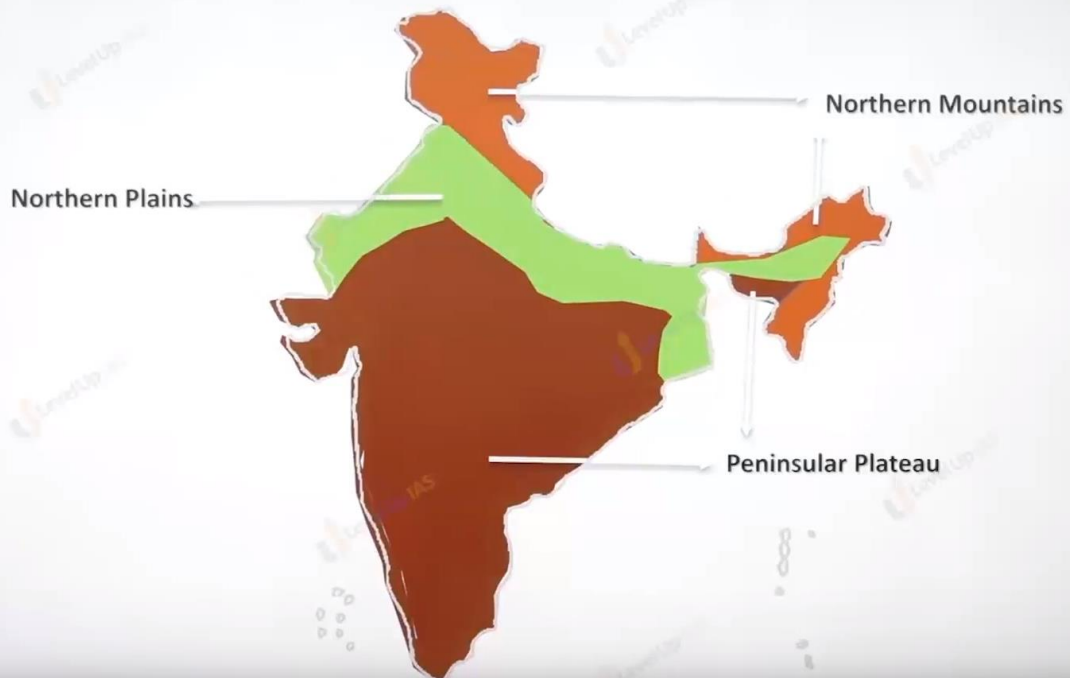
Important Lines

- Tropic of cancer passes through which states?
- IST passes through which states?
- North, South, West, East point of India

-
1. Along the longitude of Delhi you have Shimla Bhopal Bangalore Kanyakumari.
-

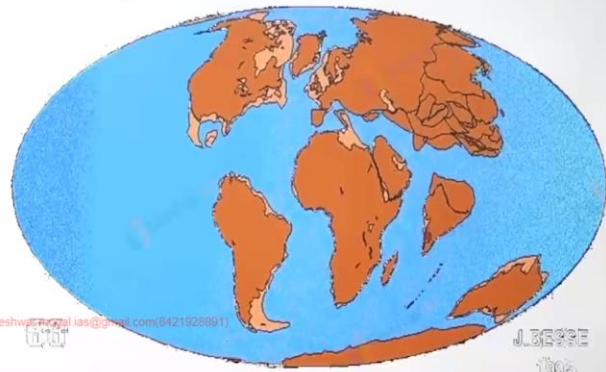


Triple tectonic division of India



History of India:

1. Indian plate was to the south of the equator millions of years ago
 2. It was much larger in size and the Australian plate was a part of it.
 3. Over millions of years, this plate broke into many parts and the Australian plate moved towards the south-eastern direction and the Indian plate to the north.
- Northward movement of the Indian plate is still continuing



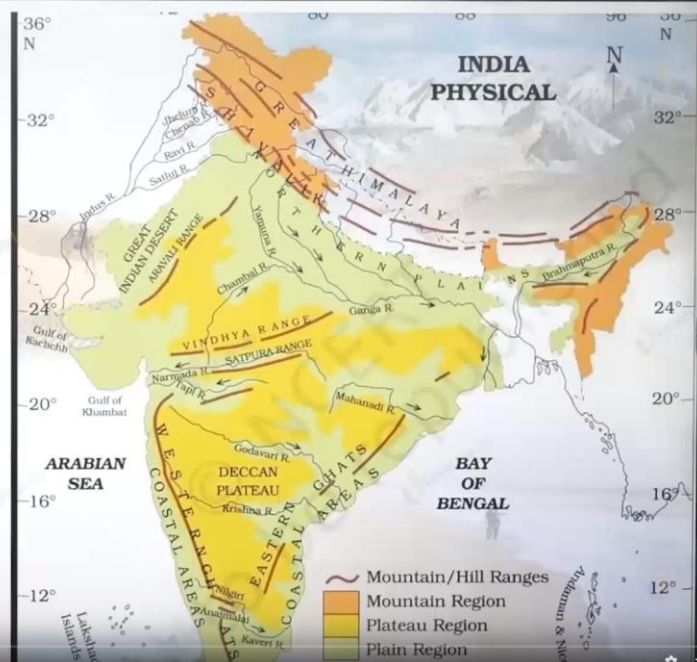
About the Peninsular Block: Shape and Extent

1. **Shape:** Northern boundary is an irregular line running from Kachchh following the Aravali Range then roughly parallel to the Yamuna- Ganga as far as the Rajmahal Hills and the Ganga delta. Southern: irregular triangle
2. **Karbi Anglong and Meghalaya Plateau** in the NE is part of Plateau. NE parts are separated by the Malda fault in West Bengal from the Chotanagpur plateau
3. **Rajasthan** also has extensions of block. In Rajasthan, desert overlay this block.



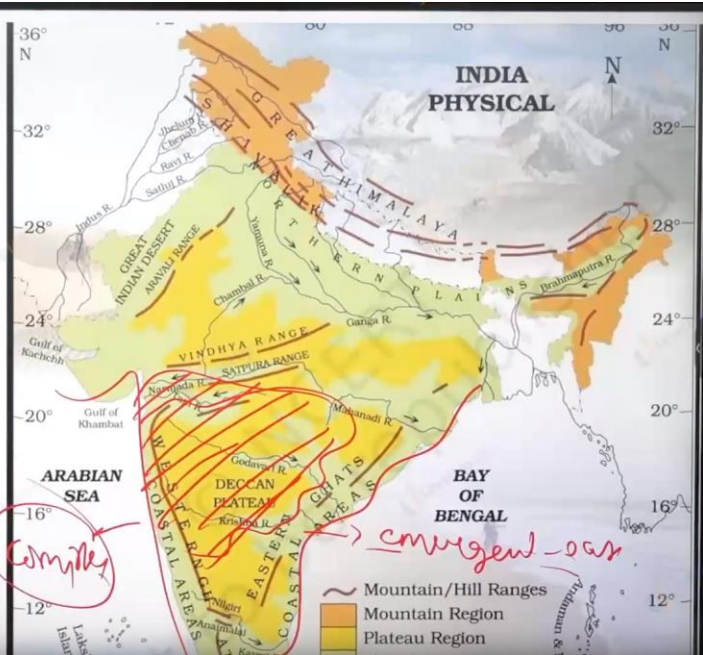
Peninsular block

1. **Height:** 150 m-900 m
2. **Mountains on outer extension:** Delhi ridge in the northwest (extension of Aravalis), Rajmahal hills in the east, Gir range in the west and the Cardamom hills in south
3. Series of plateaus such as the **Hazaribagh plateau, Palamu plateau, Ranchi plateau, Malwa plateau, Coimbatore plateau and Karnataka plateau.**



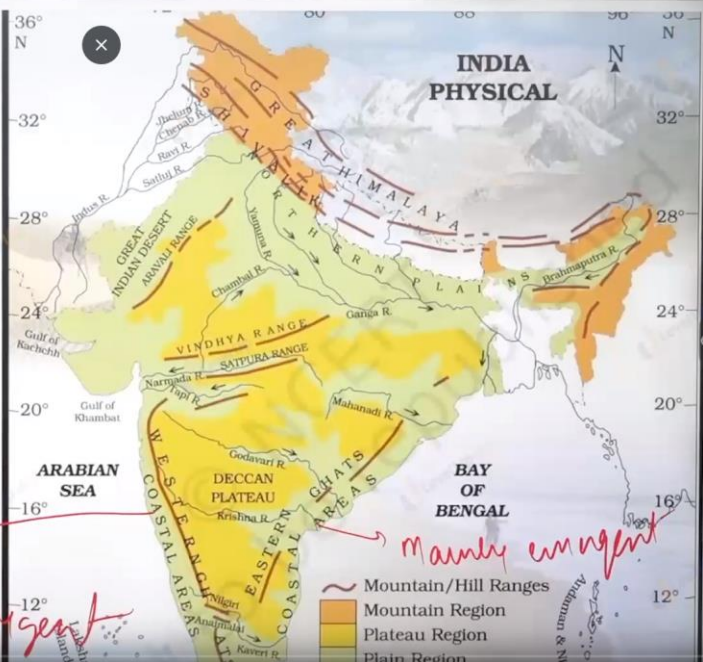
Peninsular block

1. General elevation of plateau: West to East, proved by the pattern of the flow of rivers
2. Western and North-western part of Plateau has black soil.
3. Peninsular plateau has undergone through **phases of upliftment and submergence**



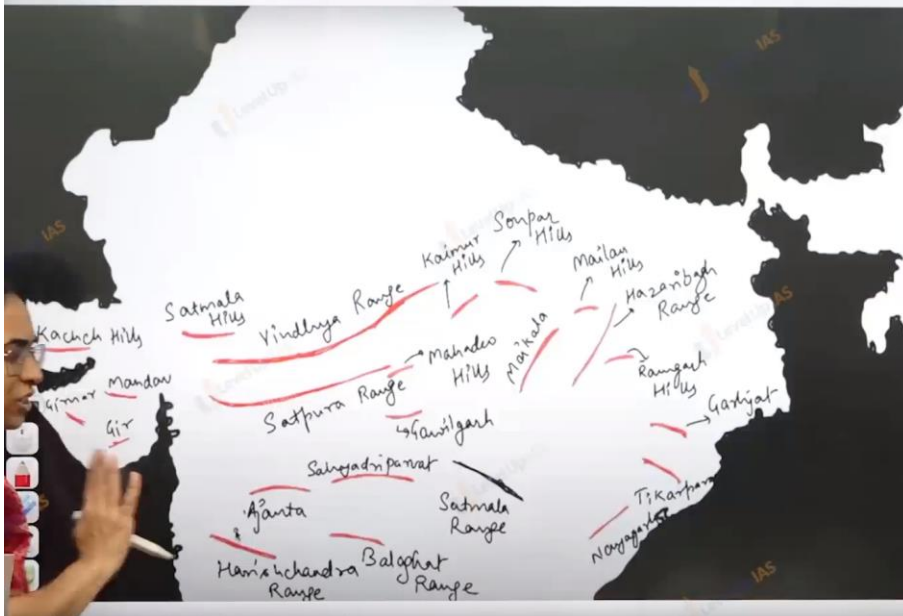
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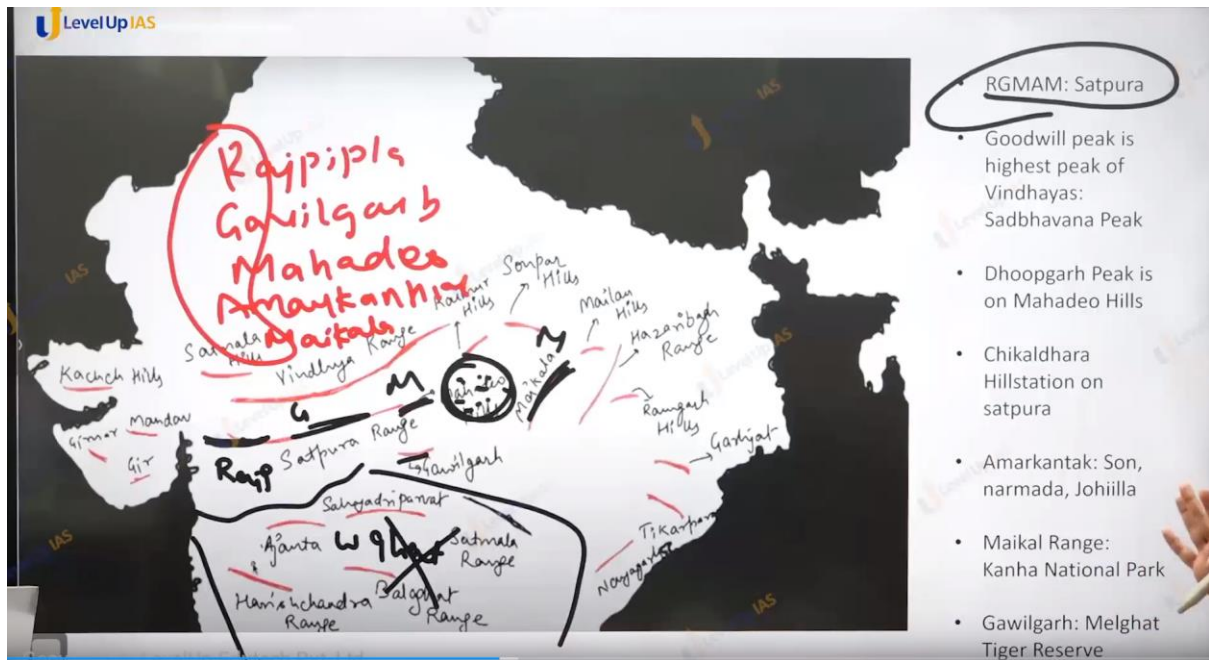


Central Highland

1. Bounded to west by **Aravali range**.
2. **Satpura range** forms the southernmost part of Central Highlands. These are **relict mountains** which are highly denuded and form discontinuous ranges.
3. **Slope of Central Highlands**: From south-western to north-eastern directions .
4. **Origin of rivers**: Most of tributaries of Yamuna have their origin in the Vindhyan and Kaimur ranges.
(Exception: Banas, tributary of Chambal originates from Aravalli)
5. **Eastern extension** of Central Highland is **Patkai Bhal hills**.
6. South-Eastern side: **Chotanagpur Plateau**

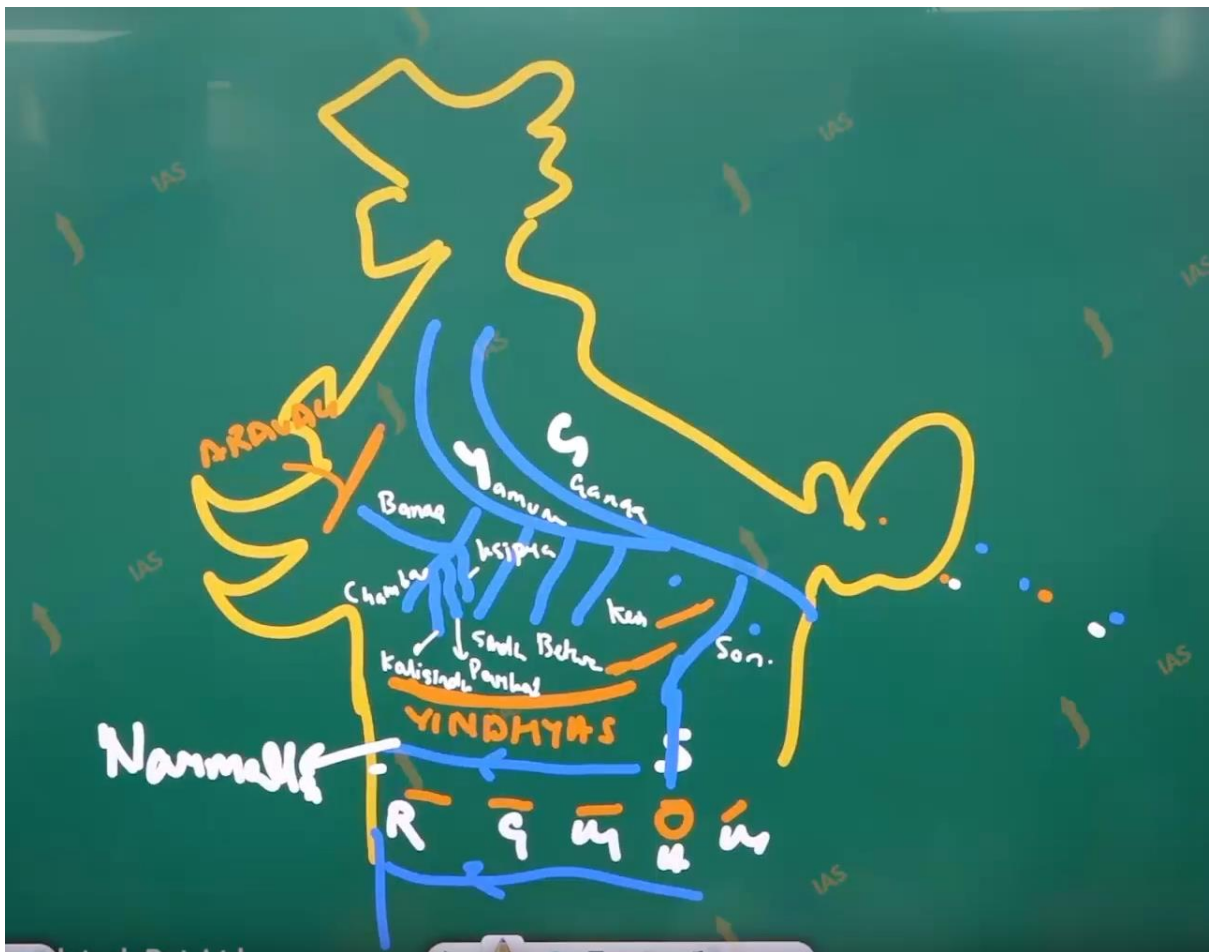


- RGMAM: Satpura
- Goodwill peak is highest peak of Vindhayas: Sadbhavana Peak
- Dhoopgarh Peak is on Mahadeo Hills
- Chikaldhara Hillstation on satpura
- Amarkantak: Son, narmada, Johiilla
- Maikal Range: Kanha National Park
- Gawilgarh: Melghat Tiger Reserve



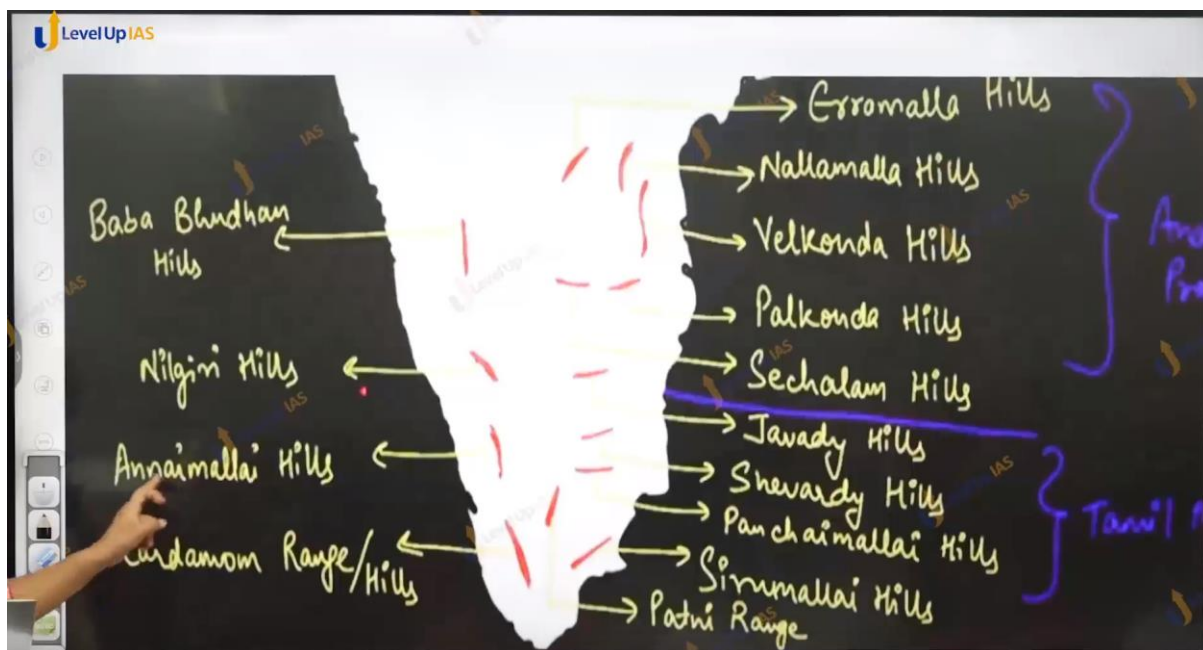
Central highland

1. The direction of reiver like Chambal Sind Kent.
2. Bounded to west by Aravalli range
3. And origin of most of river here is in the central island. And most of the tributaries of Chambal also has there origin in the vindidyas.
4. Except Banas.



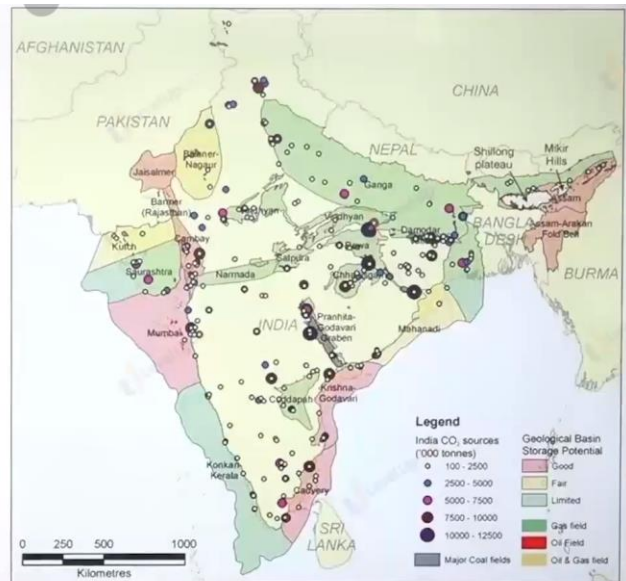
Deccan Plateau

- **Bordered by the**
 - Western Ghats in the west,
 - Eastern Ghats in the east
 - Satpura range in the north.
- Western Ghats are **locally** known as such as
 - **Sahyadri** in Maharashtra
 - **Nilgiri hills** in Karnataka -Tamil Nadu
 - **Anaimalai hills** and **Cardamom hills** in Kerala.



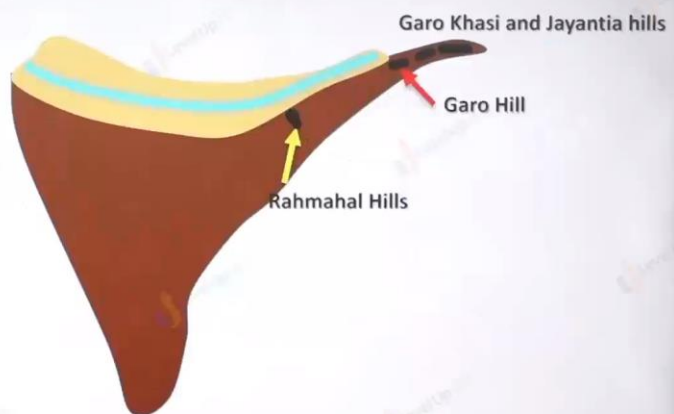
Marine Transgression

- As India Plate moved northward, there are 4 events of marine transgression (sea water entering into Peninsular Plateau)
- These areas form the major oil and gas resources
 - Kutch Saurashtra Shelf: South Bassein, Bombay High
 - K G Basin: Rawa Oil Fields
 - Assam Region: Naharkatiya, Bhasmarg, Moran Hugrijan.



North Eastern Plateau

- Extension of Peninsular plateau.
- Force exerted by the northeastward movement of the Indian plate at the time of the Himalayan origin, a huge depression was created between the Rajmahal hills and the Meghalaya plateau.
- Depression created is called as Rajmahal Garo Gap
- Later, this depression got filled up by the deposition activity of the numerous rivers.
- Today, the Meghalaya and Karbi Anglong plateau stand detached from the main Peninsular Block.



Peninsular Block: Movements Experienced

1. **Composed:** Ancient gneisses and granites. Hard Rocks and so stable
2. One of the **oldest and the most stable landmass** of India Since the **Cambrian period**, the Peninsula has been standing like a rigid block in stable condition.
3. Peninsula has relict-residual mountains like **Aravali, Nallamala, Javadi, Veliconda, Palikonda, Mahendragiri hills**
4. But it has gone tectonic movements like:
 1. Formation of Western Ghat
 2. Rifting
 3. Formation of Deccan Lava Plateau
 4. Formation of Rajmahal Garo Gap
 5. Marine Transgression

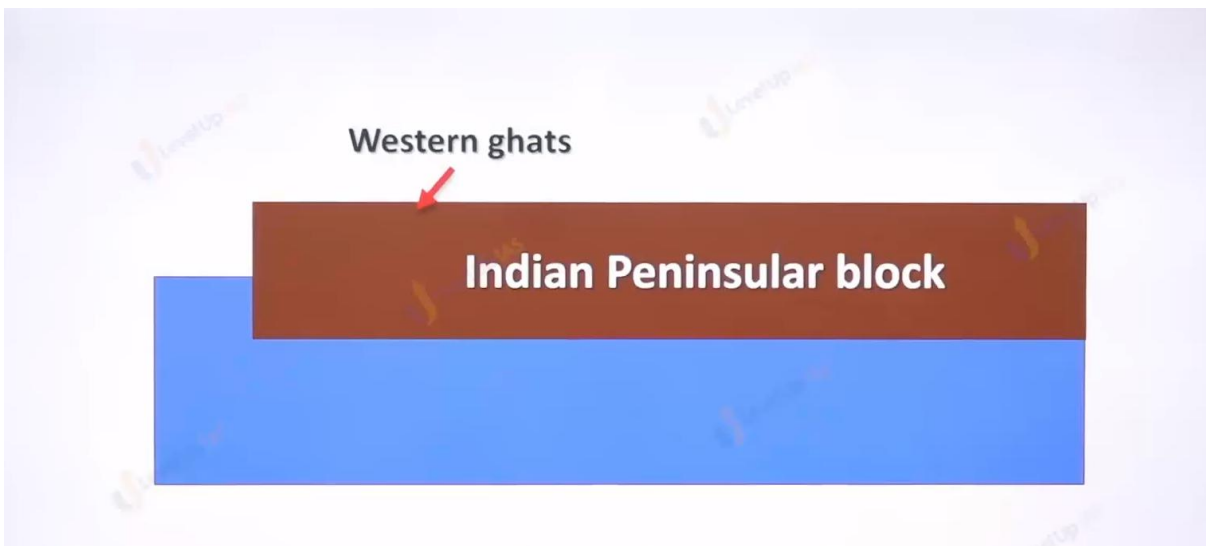


Peninsular Block :

1. They are composed of Ancient genesis of granites hard rock so stable.
2. This Cambrian period.

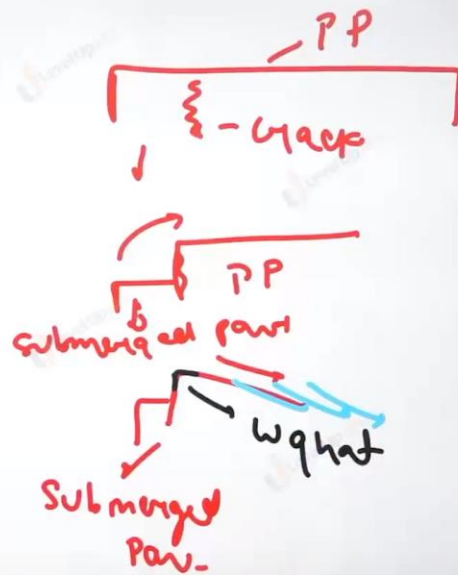
Formation of Western Ghat

1. Peninsular Plateau was bigger in size than present day size
2. Peninsular Plateau developed cracks and the western side of peninsular plateau submerged under water
3. Remaining portion of peninsular plateau rotated eastward and western edge of peninsular plateau formed Western Ghat
4. Western Ghats are block Mountains

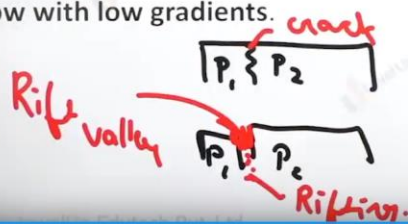


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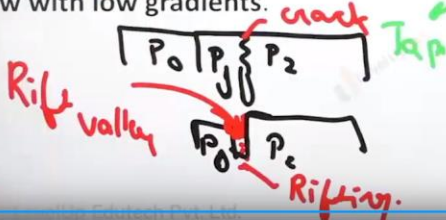
1. Due to Plate movement, PP has undergone **vertical movements** like faulting, rifting.
2. Eg: Rift valleys of Narmada, Tapi, Mahanadi, Son, Damodar, Subarnarekha. Rivers flow via these rift valleys and these are associated with the coal fields.
3. River valleys in peninsular India are shallow with low gradients.



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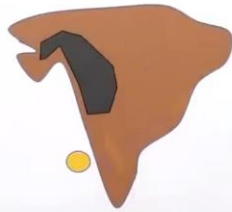


Eurasian plate

Reunion Hot Spot

Eurasian plate

Reunion Hot Spot



Eurasian plate

Indian plate

Treppen

Deccan trap

BP / CP → Deccan trap (Plutonic)

Hot Spot



Formation of Deccan lava Plateau

Indian Plate was passing over Reunion Hot Spot

Western Part experienced lava flow over it

It formed deccan lava Plateau

Location: Maharashtra, Gujarat, Some part of M.P, Malwa.

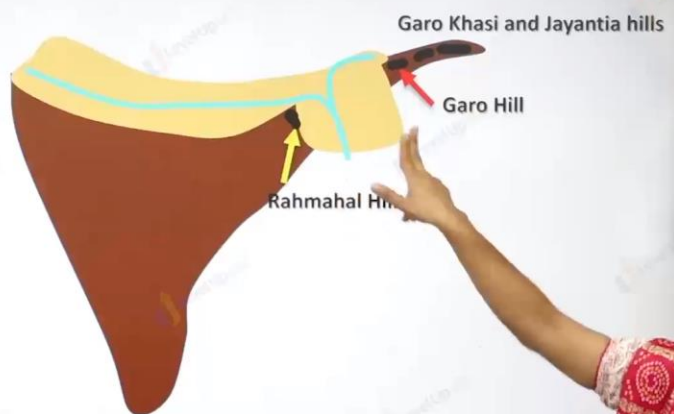
Region has Basalt rocks and forms the regur soil of India

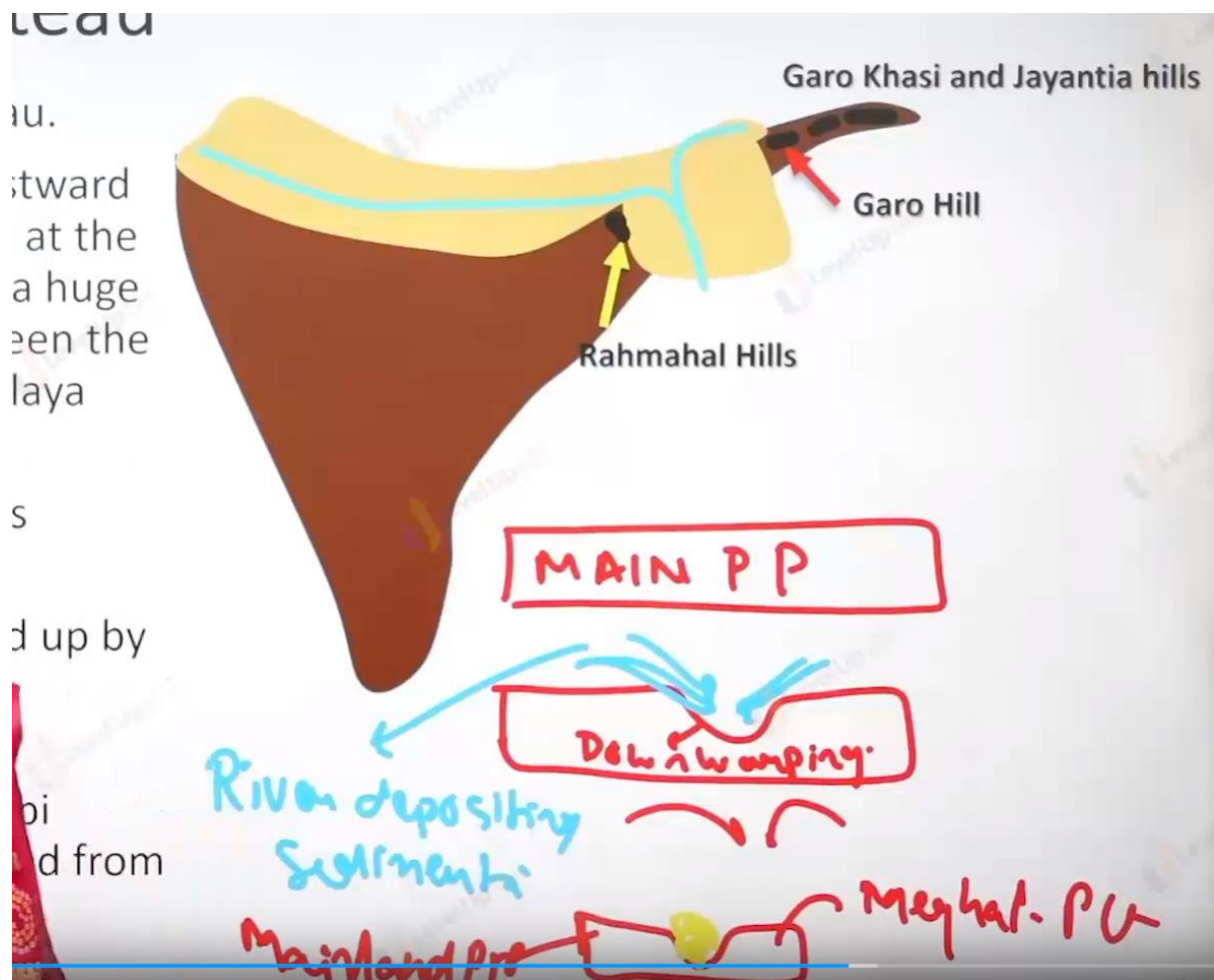
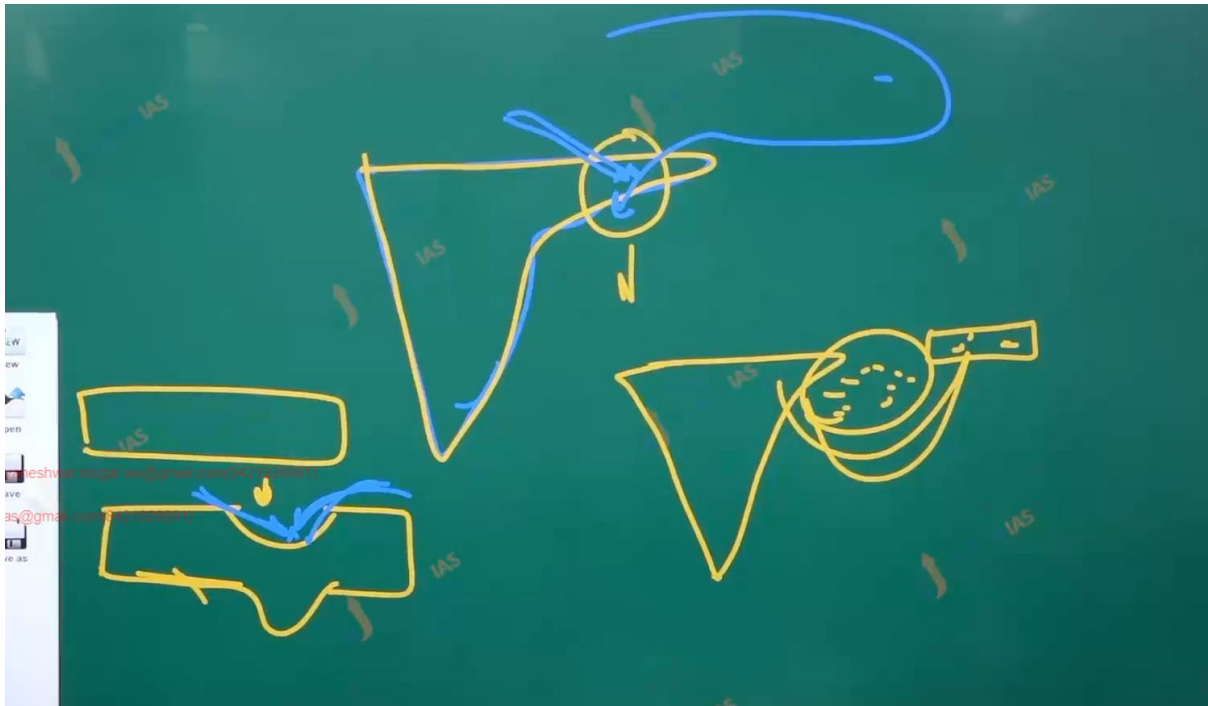


1. The western part of India That experienced lava flows was in the formed of services of flow forming step like structure (trepan) therefore this regions also called as decan lava trap.
2. Back cotton soil. Are found.
- 3.

North Eastern Plateau

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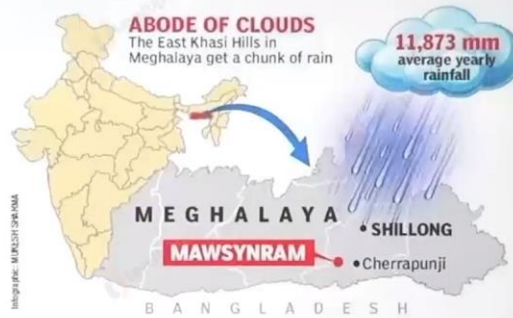




North Eastern Plateau: Meghalayan Plateau

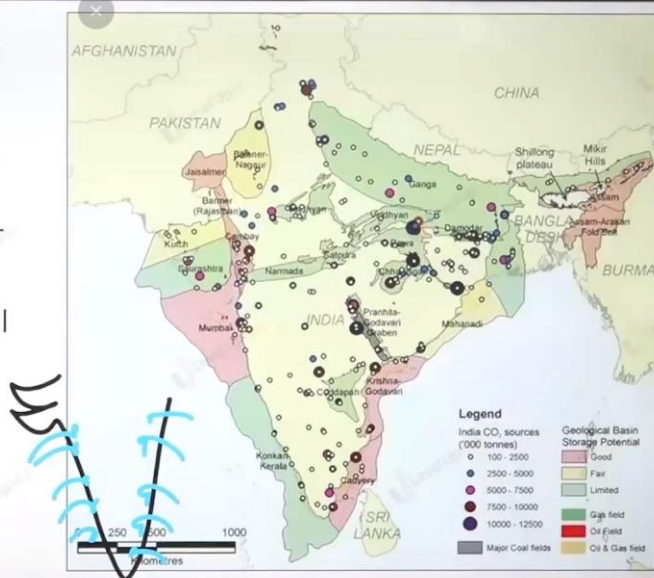
- Meghalaya plateau is further sub-divided into three:
 - (i) The Garo Hills;
 - (ii) The Khasi Hills;
 - (iii) The Jaintia Hills
- Extension of this is also seen in Karbi Anglong hills of Assam.
- Similar to the Chotanagpur plateau, the Meghalaya plateau is also rich in mineral resources like coal, iron ore, sillimanite, limestone and uranium.

This area receives maximum rainfall from the south west monsoon. As a result, the Meghalaya plateau has a highly eroded surface. Cherrapunji displays a bare rocky surface devoid of any permanent vegetation cover.



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Deccan Plateau: Western Ghat

- W. Ghats has higher elevation and are more **continuous** than E. Ghats.
- Western Ghats: Elevation is 1,500 m with the height **increasing from north to south**.
- **Anaimudi**= Highest peak of Peninsular plateau on Anaimalai hills
- **Dodabetta** is highest peak of Nilgiri hills.
- **Peninsular rivers** have their **origin** in Western Ghats.



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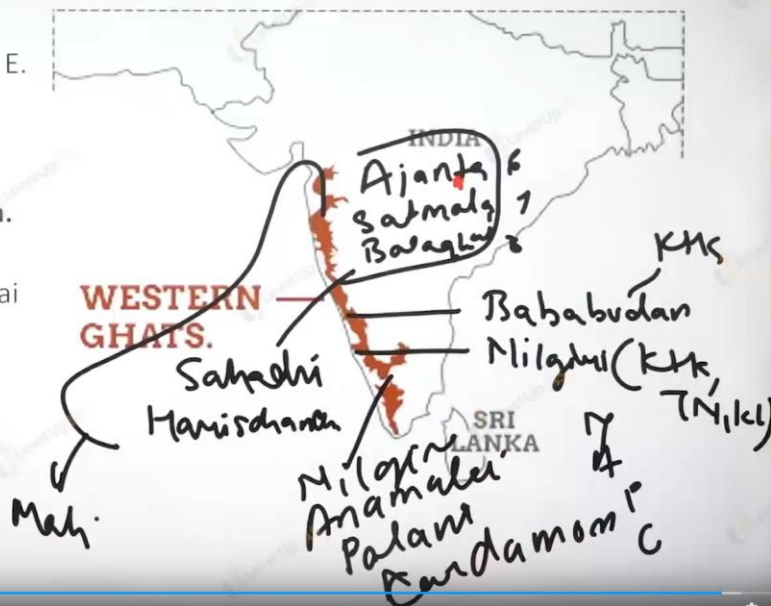
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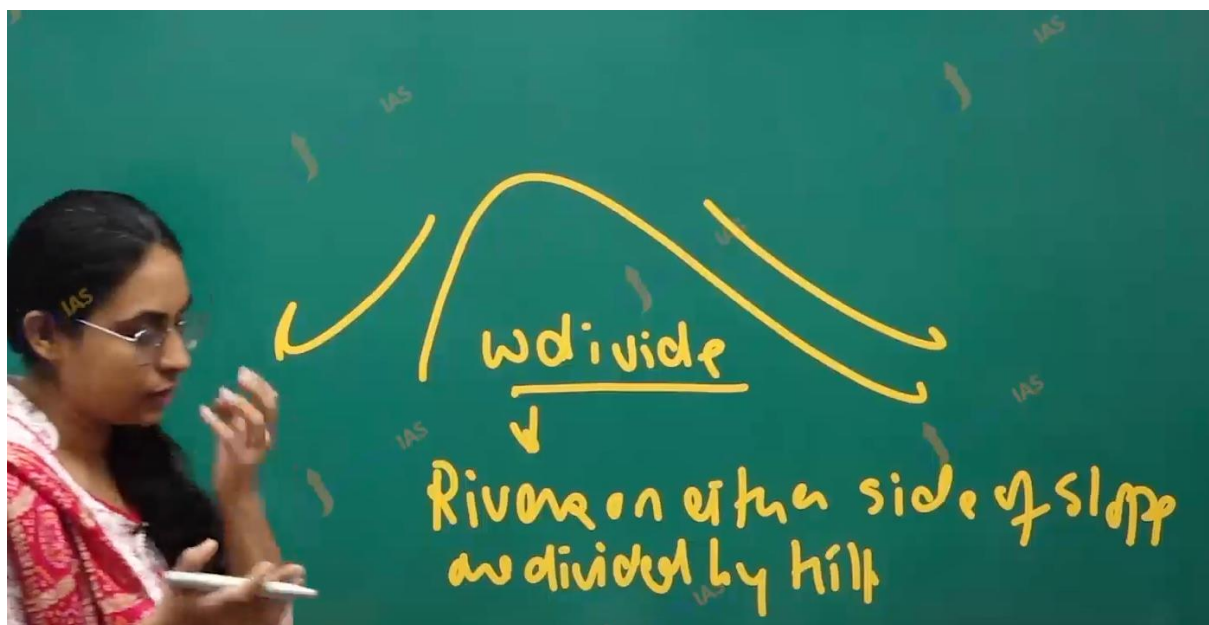
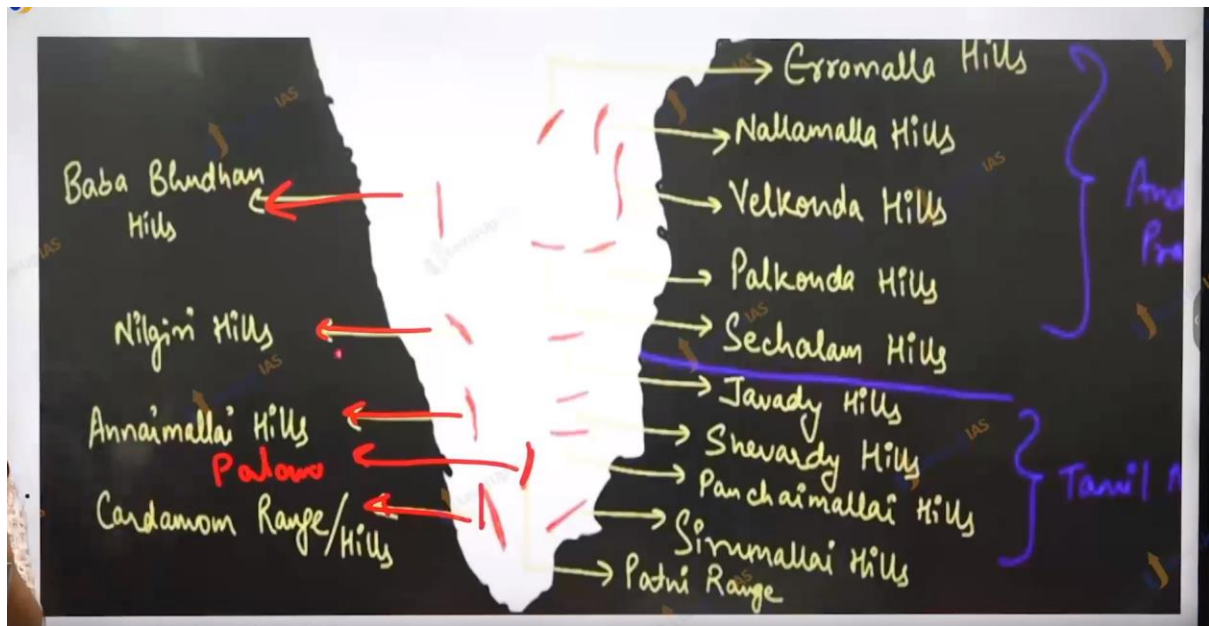
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Deccan Plateau: E Ghat

- Eastern Ghats are **discontinuous** and low hills are highly eroded by rivers of Mahanadi, Godavari, Krishna, Kaveri
- **Some of the hills** are Javadi hills, Palconda range, Nallamala hills, Mahendragiri hills, etc.
- **Eastern and Western Ghats** meet each other at the Nilgiri hills.



Ranges of Himalaya

